

Entity-Relationship (ER) Model Design

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1. INTRODUCTION

Amazon is a large-scale e-commerce platform that manages customers, products, orders, payments, and deliveries. This document presents the comprehensive system design using entities, attributes, primary keys, foreign keys, and detailed relationship mappings to establish a robust backend relational database structure.

2. ENTITIES AND ATTRIBUTES

2.1 Customer

Attribute Name	Description
customer_id PK	Unique ID for each customer
name	Customer full name
email	Customer email address
phone	Contact phone number
address	Customer primary shipping address

2.2 Product

Attribute Name	Description
product_id PK	Unique ID for product
product_name	Name/title of the product
price	Unit price of the product
stock	Available quantity in inventory
category	Product category classification

2.3 Seller

Attribute Name	Description
seller_id PK	Unique ID for seller
seller_name	Business or seller name
contact	Seller contact details

2.4 Order

Attribute Name	Description
order_id PK	Unique ID for order
order_date	Date and time of order placement
total_amount	Total monetary price for order
customer_id FK	Links to Customer entity

2.5 Order Item

Attribute Name	Description
order_item_id PK	Unique ID for individual line item
order_id FK	Links to parent Order entity
product_id FK	Links to Product entity
quantity	Number of units ordered
price	Price of item at time of purchase

2.6 Payment

Attribute Name	Description
payment_id PK	Unique ID for transaction
order_id FK	Links to associated Order entity
payment_method	Mode of payment (UPI / Credit Card / Debit Card)
payment_status	Current status (Paid/Pending/Failed)

2.7 Shipment

Attribute Name	Description
shipment_id PK	Unique ID for shipment tracking
order_id FK	Links to associated Order entity
delivery_date	Expected or actual delivery date
status	Current status (Processing / Shipped / Delivered)

2.8 Review

Attribute Name	Description
review_id PK	Unique ID for review entry
customer_id FK	Links to reviewing Customer
product_id FK	Links to reviewed Product
rating	Rating score given (e.g., 1 to 5 stars)
comment	Detailed customer feedback text

3. RELATIONSHIPS

Entity 1	Relationship	Entity 2	Cardinality / Type
Customer	places	Order	One-to-Many
Order	contains	Order_Item	One-to-Many
Product	included in	Order_Item	One-to-Many
Seller	sells	Product	One-to-Many
Order	has	Payment	One-to-One
Order	has	Shipment	One-to-One
Customer	writes	Review	One-to-Many
Product	receives	Review	One-to-Many

4. CONCLUSION

The table-based ER model clearly defines the structural architecture of the Amazon E-Commerce System. By defining precise entity schemas, primary and foreign key constraints, and relational cardinalities, this model facilitates easy understanding, optimal relational database normalization, scalable database design, and efficient transactional data management.